A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

5. **Finding a Supremum**

Consider the set

$$A = \left\{ 1 - \frac{1}{n} \;\middle|\; n \in \mathbb{N}, n \neq 0 \right\}.$$

(a) Draw a picture of A on the real line.

(b) Using your picture, determine for which values of β we can say that β is an upper bound for A in $\mathbb{R}$?

(c) Based on your answer to (b), what do you think is the least upper bound (i.e., supremum) of A in $\mathbb{R}$?

(d) Assume that your answer to (c) is incorrect. This means there is a upper bound, r, smaller than the one you have conjectured in (c). Prove there is an element of A that is bigger then r. At some point in your proof, you might need to use the following true statement:

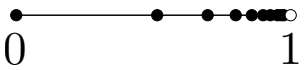
$$(\forall \epsilon > 0) \;\; [(\exists n \in \mathbb{N})\; 1/n < \epsilon]$$

(Hint: It might help to draw a picture. Based on the picture, what do you want to show?)

(e) Does your work in the previous part convince you that your answer in (*c*) is correct? Take a moment to refer back to the definition of <u>supremum</u> to be certain.

Solutions:

(a)



(b) $\beta \geq 1$

(c) $\beta = 1$

(d) Assume $r < 1$. Let $d = 1 - r$. There exists $n \in \mathbb{N}$ so that $\frac{1}{n} < d$. Therefore

$$r = 1 - d < 1 - \frac{1}{n} < 1$$

Since $1 - \frac{1}{n} \in A$ and $r < 1 - \frac{1}{n}$, r is not an upper bound for A in $\mathbb{R}$.

(e) Yes. From (d), we know that an upper bound for A must be ≥ 1. Since 1 is an upper bound for A, 1 is the least upper bound of A. That is, 1 is the supremum of A.

6. **Maximums and Minimums**

Consider the following definitions:

Definition. *Let S be an ordered set, let $A \subseteq S$ such that A has an supremum in S, and let $m \in A$. We say <u>m is maximum in A</u> when* $\sup A = m$.

Definition. *Let S be an ordered set, let $A \subseteq S$ such that A has an infimum in S, and let $m \in A$. We say <u>m is minimum in A</u> when* $\inf A = m$.

Find, if they exist, the supremum, infimum, maximum and minimum of each set A. Consider each set as a subset of $\mathbb{R}$. You do not need to prove your answers are correct, but certainly you should be able to explain verbally why your answers are correct.

(a) $A = \{x \in \mathbb{R} \mid x^2 \leq 3\}$.

Solution: $\inf A = \min A = -\sqrt{3}$, $\sup A = \max A = \sqrt{3}$.

(b) $A = \{x \in \mathbb{Q} \mid x^2 \leq 3\}$.

Solution: $\inf A = -\sqrt{3}$, $\sup A = \sqrt{3}$. Since $\sqrt{3}$ is irrational, this set has no maximum nor minimum.

(c) $A = \{x \in \mathbb{R} \mid x \leq \pi\}$.

Solution: This set has no infimum (hence no minimum). $\sup A = \max A = \pi$.

(d) $A = \{x \in \mathbb{Q} \mid x \leq \pi\}$.

Solution: This set has no infimum (hence no minimum). $\sup A = \pi$. Since π is irrational, this set has no maximum.

(e) $A = [-1, 1)$.

Solution: $\inf A = \min A = -1$, $\sup A = 1$. This set has no maximum.

(f) $A = (-1, 1)$.

Solution: $\inf A = -1$, $\sup A = 1$. This set has no maximum nor minimum.

When we from $\mathbb{R}$ to $\mathbb{Q}$ in (b) and (d), the sets no longer have a supremum nor infimum in $\mathbb{Q}$.

7. **Additive Inverses are Unique**

Let F be a field and let x be an element of F.

(a) Let y and y' be additive inverses of x in F. What does the definition of <u>additive inverse</u> tell us about the expressions $x + y$ and $x + y'$?

(b) Using your answer from the previous part and the statement of Theorem 2.23, what can you conclude about y and y'?

(c) Does the name of this question make sense?

Solutions:

(a) $x + y' = 0$, $x + y = 0$

(b) They are equal.

(c) There is only one possibility for the additive inverse of each element x of a field. In other words, the the additive inverse of x is unique.

8. **Additive Identity is Unique**

(a) In part (a) of the previous question you stated that both $x+y$ and $x+y'$ were equal to the additive identity of F. Does the additive identity axiom say that there is only one additive identity in a field?

(b) Using the Field Axioms, prove that the 0 and $0'$ are both additive identities in a field, then $0 = 0'$. (Hint: What do you get if you add these two elements together?)

(c) Does the name of this question make sense?

Solutions:

(a) It does not.

(b) By (A4) $0 + 0' = 0$ and $0' + 0 = 0'$. By (A2) $0 + 0' = 0' + 0$. Therefore $0 = 0'$

(c) There is only one possibility for the additive identity of an element of a field. In other words, the the additive identity is unique.

9. **Multiplicative Identity is Unique**

Using the Field Axioms, prove that the multiplicative identity in a field is unique.

Solution: Let 1 and $1'$ both have fulfill the properties of the multiplicative identity. By axiom (M4), we have $1 \times 1' = 1'$ and $1' \times 1 = 1$. By axiom (M2), we have $1 \times 1' = 1' \times 1$. Therefore, $1' = 1$. In other words, the multiplicative identity is unique.